MOTION

Class 9 | Science | Chapter 8

MOTION & REST

An object is in motion if its position changes with time w.r.t. a reference point (origin); otherwise it is at rest.

Motion is relative -- depends on the observer/reference point chosen.

DISTANCE & DISPLACEMENT

Term	Meaning
Distance	Total path length covered -- always positive, scalar quantity
Displacement	Shortest straight-line distance from initial to final position, with direction -- vector quantity

Displacement can be zero even when distance travelled is not zero (e.g. one full round of a circular track).

SCALAR VS VECTOR

Scalar (magnitude only)	Vector (magnitude + direction)
Distance, Speed, Mass, Time	Displacement, Velocity, Acceleration, Force

SPEED, VELOCITY & GRAPHS

Uniform & Non-Uniform Motion

SPEED & VELOCITY

Speed = Distance / Time (scalar). Velocity = Displacement / Time (vector, has direction).

Uniform Motion

Equal distances covered in equal intervals of time -- speed remains constant.

Non-Uniform Motion

Unequal distances covered in equal intervals of time -- speed changes.

Average Speed

Average speed = Total distance / Total time.

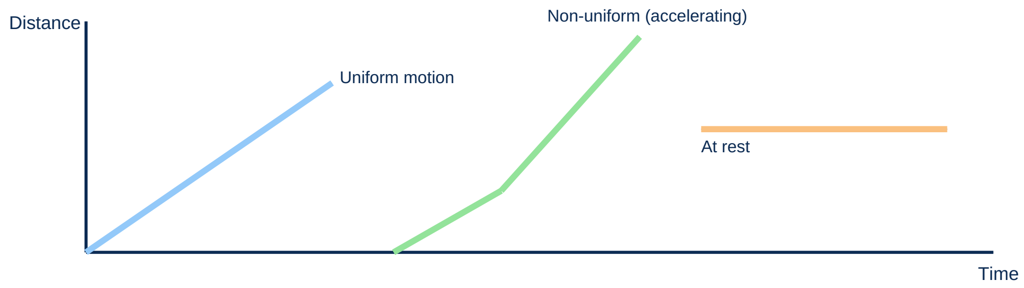
ACCELERATION

Acceleration = Change in velocity / Time taken = (v - u) / t

Positive acceleration: speeding up. Negative acceleration (retardation/deceleration): slowing down. SI unit: m/s^2.

DISTANCE-TIME GRAPHS

DIAGRAM: DISTANCE-TIME GRAPH PATTERNS



Straight sloped line = uniform motion. Curved line = non-uniform (accelerated) motion. Horizontal line = object at rest.

EQUATIONS OF MOTION

Formulae, Circular Motion

THREE EQUATIONS OF MOTION

$v = u + at$

$s = ut + (1/2)at^2$

$v^2 = u^2 + 2as$

Where u = initial velocity, v = final velocity, a = acceleration, t = time, s = distance travelled.

VELOCITY-TIME GRAPHS

Graph shape	Meaning
Horizontal line	Uniform velocity (zero acceleration)
Straight line, sloping up	Uniform acceleration
Straight line, sloping down	Uniform retardation (deceleration)
Area under v-t graph	Gives the distance/displacement covered

UNIFORM CIRCULAR MOTION

Motion in a circular path at constant speed -- velocity keeps changing direction, so it is an accelerated motion.

Ex: Motion of the tip of a clock's second hand; a satellite orbiting Earth.

Circumference of a circle = 2 x pi x r (used to calculate speed in circular motion).

SUMMARY & REVISION

Key Formulae + Quick Recap

CHAPTER SUMMARY -- KEY TERMS

- **Distance** -> path length (scalar); **Displacement** -> shortest path with direction (vector)
- **Speed** -> distance/time; **Velocity** -> displacement/time
- **Acceleration** -> rate of change of velocity
- **Uniform motion** -> equal distances in equal times
- **3 equations of motion** -> $v=u+at$, $s=ut+1/2at^2$, $v^2=u^2+2as$

SOLVED EXAMPLE

Q: A car starts from rest and accelerates at 2 m/s^2 for 5 s. Find its final velocity.

A: $u = 0$, $a = 2\ m/s^2$, $t = 5$ s. Using $v = u + at$: $v = 0 + 2(5) = 10$ m/s.

ONE-LINE RECAP FOR REVISION

Motion is relative; distance is scalar, displacement is vector.

Acceleration = rate of change of velocity.

Area under v-t graph = distance travelled.

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